

Prompt Engineering Master Handbook v3.0

Complete Reference for GenAI & AI Engineer Preparation

Part 1: Core Prompt Formula

- Role
- Task
- Context
- Goal
- Instructions
- Reasoning Process
- Constraints
- Evaluation Criteria
- Output Format

Lesson 1: Prompt Structure

- Workflow: Role → Task → Context → Output → Constraints → Response
- Purpose: Create clear, predictable prompts.

Lesson 2: Context Layering

- Keywords: Context, Background, User Profile, Goals, Skill Level, Resources, Limitations
- Workflow: User Goal + Skills + Time + Preferences → Personalized Response

Lesson 3: Few-Shot Prompting

- Provide examples before asking the task.
- Workflow: Examples → Pattern Recognition → Consistent Output

Lesson 4: Chain-of-Thought

- Guide reasoning through steps.
- Workflow: Analyze → Compare → Evaluate → Recommend

Lesson 5: Structured Outputs

- JSON
- XML
- Markdown
- Tables
- Schemas
- Workflow: Task → Schema → Machine Readable Output

Lesson 6: Constraints & Delimiters

- Constraints: Always, Never, Must, Must Not
- Delimiters: Triple Quotes, XML Tags, Code Blocks
- Workflow: Instructions + Data Separation → Controlled Output

Lesson 7: System vs User Prompts

- System Prompt = Permanent Behavior
- User Prompt = Current Task
- Workflow: System → Context → User Query → Response

Lesson 8: Prompt Debugging

- Identify Role Problems
- Identify Context Problems
- Identify Output Problems
- Optimize and Retest

Lesson 9: Evaluation & Critique Loops

- Generate → Evaluate → Critique → Improve → Final Output
- Use scoring rubrics and review frameworks.

Lesson 10: RAG Prompting

- Use Retrieved Context
- Ground answers in evidence
- Avoid assumptions
- Workflow: Retrieval → Context Injection → Answer

Lesson 11: Agentic Prompting

- Goal
- Plan
- Execute
- Evaluate
- Improve
- Workflow: Goal → Plan → Action → Review → Result

Professional Workflows

- Prompt Structure Workflow
- Context Layering Workflow
- Few-Shot Workflow
- CoT Workflow

- Structured Output Workflow
- RAG Workflow
- Agent Workflow
- Evaluation Workflow

Top 20 High-Value Keywords

- Role
- Task
- Context
- Goal
- Instructions
- Constraints
- Requirements
- Reasoning Process
- Evaluate
- Analyze
- Assess
- Rank
- Score
- Critique
- Improve
- Workflow
- Output Format
- Always
- Never
- Background

Production Prompt Templates

- Career Advisor Template
- Resume Reviewer Template
- Project Evaluator Template
- RAG Assistant Template
- AI Agent Template

Prompt Debugging Checklist

- Is the role specific?
- Is context sufficient?
- Are constraints clear?
- Is output structured?

- Are evaluation criteria defined?

Hallucination Reduction Checklist

- Use only provided context
- Require evidence
- Disallow assumptions
- Add fallback response
- Use evaluation loops

GenAI Project Prompt Patterns

- AI Resume Analyzer
- AI Career Mentor
- RAG Chatbot
- Research Assistant
- Multi-Agent Workflow

Interview Preparation

- Explain RAG
- Explain Agents
- Explain Structured Outputs
- Explain Evaluation Loops
- Discuss Production Projects

Master Production Architecture

• System Prompt → Context Layering → Retrieval → Examples → Reasoning → Constraints → Structured Output → Evaluation → Critique → Final Response